Formality as a Financing Decision

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The link between informality and financial conditions

Informality:

- Prevalent across time and regions.
- Implications on tax collection, inequality, etc.
- Failed policies, e.g. special tax regimes (RIF, REPECO, etc).

Informal: An establishment with positive income and no Social Security contributions.

▸ Informality transitions ▸ Intensive margin ▸ IMSS employer number ▸ RFC tax number

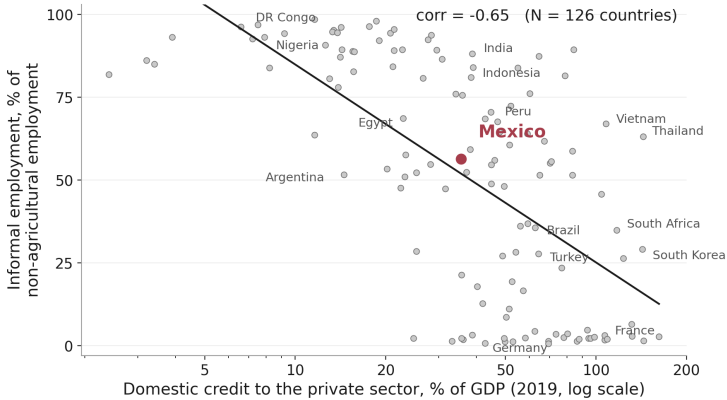
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This paper:

- Financial conditions?



Sources: ILO SDG 8.3.1 via Our World in Data; World Bank WDI (2019).

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Contribution

- First causal evidence that bank credit supply shocks increase firm formalization.
- $\uparrow$ Credit $\Rightarrow \downarrow$ Capital constraints $\Rightarrow \uparrow$ Formal entry & formalization $\Rightarrow \downarrow$ Informality.

Outline

1) Literature
2) Data
3) Model
4) Empirical Strategy
5) Results
6) Conclusions

Literature

Credit supply effects on firm activity (Khwaja and Mian, 2008; Chodorow-Reich, 2014; Amiti and Weinstein, 2018; Greenstone et al., 2020; Gutierrez et al., 2023), etc.

- **Literature takeaway**: Credit affects only formal employment and investment.
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Informality: causes and consequences (La Porta and Shleifer, 2014; Meghir et al., 2015; Busso et al., 2012; Almeida and Carneiro, 2012; Ulyssea, 2018; Adom, 2025), etc.

- **Literature takeaway**: No firm dynamics, formality as an absorbing state.
- **Contribution**: Full dynamics, entry, exit, and switching in both directions.

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$\Rightarrow$ **This paper**: Formality is (also) a financing decision.

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Mexican data

- Matched bank–employer–employee data Social Security (IMSS) + National Banking and Securities Commission (CNBV)
  - 2005–2025: monthly credit registry, annual market panel.
  - Universe of formal employers and workers.
  - Universe of business loans to formal and informal firms.

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  - Every 5 years (2009, 2014, 2019, & 2024).
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Key feature: Credit + formality + firm dynamics observed in the same markets.

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Settings, simplified model version (two periods)

Technology and heterogeneity

- $y = \theta k^{\alpha} \ell^{\nu}$, $\alpha + \nu < 1$.
- $\ln \theta_1 = \ln v + u_1$.
  $\ln \theta_2 = \rho \ln \theta_1 + (1 - \rho) \ln v + u_2$.
  - $\Pi_s(x, k) = cxk^{\alpha/(1-\nu)}, \quad x \equiv \theta^{\frac{1}{1-\nu}}$

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- Informal, s = i: evades both.

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- Cost: $\Phi_s(a) = r_s \times a, \quad r_i > r_f$.
- Collateral: $a \leq b_s k', \quad b_i < b_f$.
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Timing: Decisions, before u_2 observed, based on expected productivity $\hat{x} = \mathbb{E}[x_2 \mid \theta_1, v]$.

Value functions

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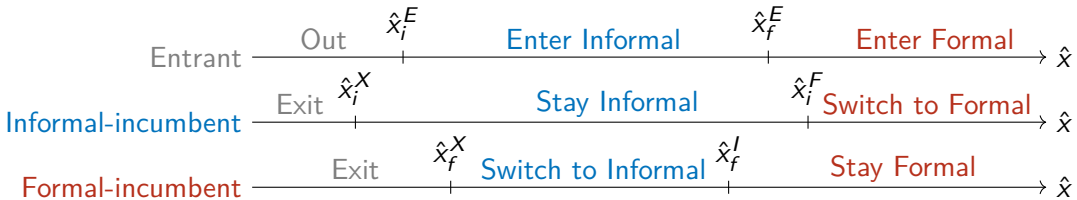
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Sector decision: Firms choose the status that maximizes lifetime value.

Unique threshold decision rules

There exists a unique threshold for each transition: $\hat{x}_i^X < \hat{x}_i^E < \hat{x}_f^X < \hat{x}_f^I < \hat{x}_f^E < \hat{x}_i^F$.



Financial conditions?: Thresholds depend on finance for constrained firms, $\hat{x}_s^S(r_s, b_s)$.

But who are the constrained firms?

▸ Formality threshold closed form ▸ Entry-Exit threshold closed form

Investment and financing: a four-region policy

$$\textbf{Optimal capital} = \begin{cases} \bar{k}_s \Leftarrow \mathrm{MPK} = 1 - \beta(1-\delta) \equiv \mathrm{MCK} & \text{if no borrowing.} \\ \tilde{k}_s \Leftarrow \mathrm{MPK} = 1 - \beta(1-\delta) + r_s \equiv \mathrm{MCK} & \text{if borrowing.} \end{cases}$$

Region	Funds (n) vs. k_s^*	Capital k_s^*	Borrowing a_s^*	Cash-flow sens.
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Predictions: Better formal loan terms lower the formality thresholds (↑ formal entry, ↑ i→f, ↓ f→i). With the output price responds, ↑ exit and no ↑ informal entry.

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Empirical strategy: Leave-One-Out Shift-Share IV, LOO-SSIV

Intuition: Each bank's credit supply growth (estimated nationally w/multi-bank firms, excluding target market) weighted by that bank's initial lending share in the target market.

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$$\Delta \ln L_{bft}^{-m} = \alpha_{ft}^{-m} + \gamma_{bt}^{-m} + \epsilon_{bft}, \qquad \Rightarrow \qquad \hat{\gamma}_{b,t}^{-m} = \text{Exogenous shift.}$$

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3) Instrument: Bank change in credit supply weighted by market share.

$$\hat{Z}_{m,t}^{LOO} = \sum_b \left(s_{b,m,t-1} \times \hat{\gamma}_{b,t}^{-m} \right)$$

Interpretation: Different exposure of markets to exogenous bank shocks.

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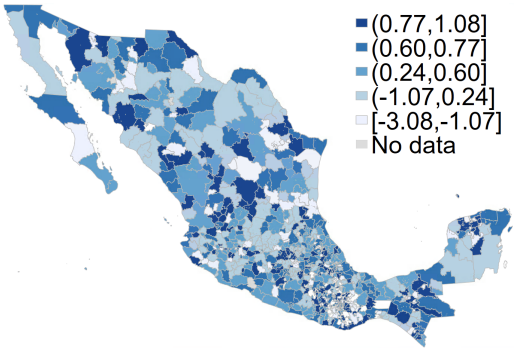
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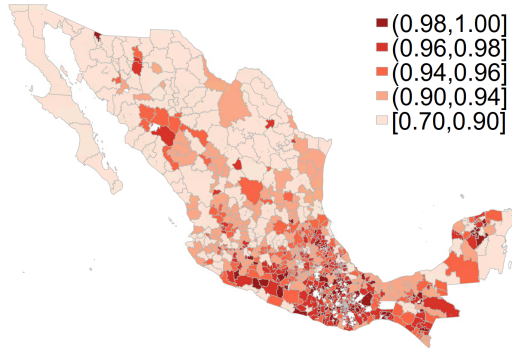
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The identifying variation across local markets



Instrument

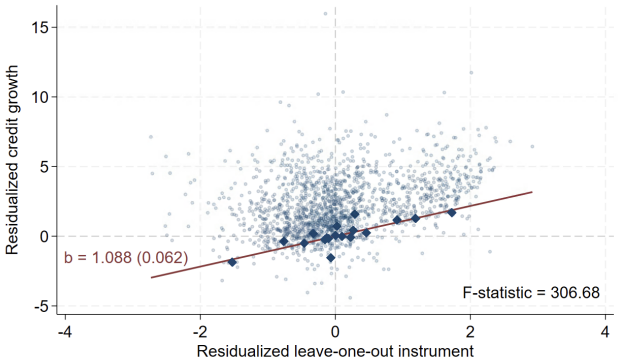


Informality share

Takeaway: IV varies across markets and it is unrelated to baseline informality.

▸ Credit change map

The instrument is relevant



- +1 s.d. shock doubles five-year credit: +70 log points (+2.5 M USD, median mkt).
- Relevant: 1st stage F = 266–290.
- 67 banks with shocks; effective sample size 16.6 (3.4–8.4 by window) ⇒ Concentrated but no single bank drives results.

Interpretation: Shocks capture supply variation, but where does this shock go?

A credit supply shock deepens more than it widens

What does a 1 s.d. annual supply shock imply?

- $+14$ log points of market credit ($\approx$15%) in the year of the shock, persistent: $+22$ at two years, $+20$ at four.
- $+375$ thousand USD in the median market (Median credit $= 2.5$ million USD): one month of formal wages, 8% of the annual formal wage bill, 1.6% of borrowers' annual revenues.

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Where does this credit go?

- Credit per loan: +20% (formal) and +11% (informal) in the year of the shock, rising through year three.
- Numbers of loans and borrowers: no effect from the second year on.
- Five years: firms w/credit +3.5% per +10% credit (+4.2% formal, +3.4% informal).

Supply shock: mostly larger loans (+10%) for existing borrowers (+3.5% new borrowers).

A credit supply shock improves loan terms more for formal borrowers

Credit (1 s.d. annual shock): formal $+19$ log points vs informal $+8$, contemporanously.

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- Formal: -0.27 pp at the shock, -0.52 pp four years later (3.4% of the median formal rate).
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Collateral per peso of credit: formal -0.29 log points at four years ($\approx -25\%$); informal -0.14 (n.s.).

- Difference not statistically significant: the shock relaxes collateral for both groups.

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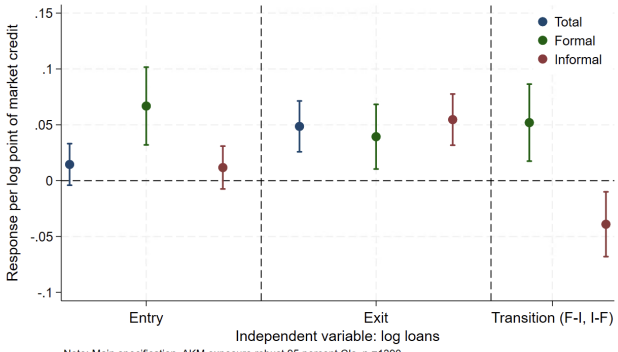
- Formal: -0.27 pp at the shock, -0.52 pp four years later (3.4% of the median formal rate).
- Informal: no response (-0.02, -0.12 pp, n.s.). The gap widens by 0.41 pp

Collateral per peso of credit: formal -0.29 log points at four years ($\approx -25\%$); informal -0.14 (n.s.).

- Difference not statistically significant: the shock relaxes collateral for both groups.

Mechanism: Better terms for formal borrowers $\Rightarrow$ the relative financial benefit of formality rises.

More credit generates firm reallocation & formal firm creation

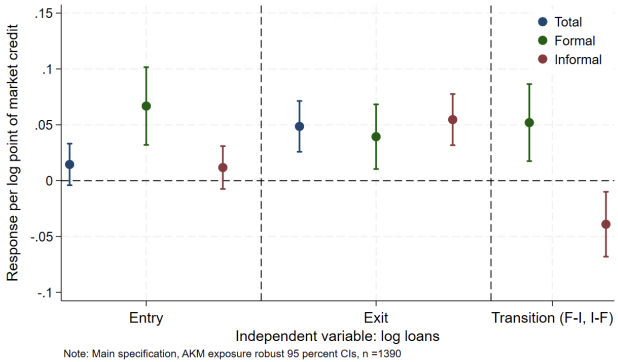


Note: Main specification, AKM exposure robust 95 percent CIs, n =1390

A supply driven ↑ 10% mkt credit:

- **Entry**: +3.1 (+1,700 national) formal firms (+0.7%).
- **Exit**: +15 (+8,300) firms (+0.5%; formal +0.4%, informal +0.6%).
- **Transitions**: +1.6 (+850) formalizations, −1.1 (−600) in-formalizations.

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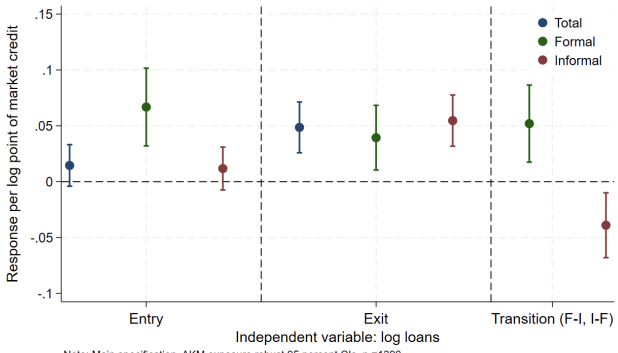


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VS tax cut: +3.5% formal firms per 1 s.d. shock vs halving Brazil's tax (+4.3%) (Rocha et al., 2018)

Credit raises formal employment & reduces informal production

	Formal labor market (IMSS)			Value added (census)			
	Workers	Wage bill	Mean wage	Avg. VA	Total	Formal	Informal
Δ_5 ln(credit)	0.020***	0.024**	−0.012***	0.029*	0.032*	0.028	−0.049**
	(0.005)	(0.009)	(0.004)	(0.015)	(0.017)	(0.031)	(0.021)
Mean dep. var.	30,967	\$0.67m	\$10.7	\$56k	\$1.18bn	\$0.88bn	\$0.30bn
Observations	1,386	1,386	1,386	1,394	1,394	1,394	1,394

Means are levels on each column's sample, in USD at 19 MXN/USD (wage bill and mean wage are daily, value added annual). Controls: employment- and credit-weighted sectoral composition, lagged market rate, log banks, log credits, uncovered share, bank share, coverage × year. AKM s.e. in parenthesis.

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composition, lagged market rate, log banks, log credits, uncovered share, bank share, coverage × year. AKM s.e. in parenthesis.

Informal VA falls with an unchanged informal stock: the most productive informal firms became formal.

Robustness checks

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9. **Different coverages**: Similar effects. Go

Consistent evidence of credit driven formalization.

Outline

1) Literature

2) Data

3) Model

4) Empirical Strategy

5) Results

6) Conclusions

Conclusions

- Documents rich firm dynamics and credit market conditions.
  - 3/4 of firms with credit are informal; informal borrowers pay 0.7 pp more, post 3 pp more collateral, get loans $< 1/3$ of formal ones.
  - Formal status is not absorbing: 22% of formal firms became informal (2019–2024).
- Causally establishes that credit supply reallocates activity toward formality.
  - Formal borrowers per 1 s.d. shock: $+19\%$ credit, -0.5 pp rates, -25% collateral; informal: $+16\%$ credit, terms unchanged.
  - $+$ formal firms, $+$ formal entry, $+$ upward switching; $-$ downward switching; $+$ exit; $-$ informal VA.
- Model's mechanism: Formality is (also) a financing decision.
  - Relative loan terms move the formality thresholds of firms that need a loan; goods market competition explains exit.
- Policy: the lever is the relative loan advantage of formal firms (credit guarantees, bank competition).
  - Sizable: 1 s.d. shock doubles credit; formal entry effect of the order of Mexico's SARE registration reform.

Thank you

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Main transition matrix

Transition matrix 2014-2019 and 2019-2024

		Destination		
		Formal	Informal	Exit
2014 Origin	Formal	122,784	23,526	56,994
	(% of Formal)	60.4	11.6	28.0
	Informal	263,565	2,232,216	1,531,660
	(% of Informal)	6.5	55.4	38.0
2019 Origin	Formal	324,052	138,041	170,946
	(% of Formal)	51.2	21.8	27.0
	Informal	146,071	2,547,652	1,473,395
	(% of Informal)	3.5	61.1	35.4

Note: Formal > 0% of workers with SS, Informal = 0% of workers with SS.

Back

Transition matrix taking into account the intensive margin

Transition matrix 2014-2019 and 2019-2024

		Destination			
		Formal	Informal	Mix	Exit
2014 Origin	Formal	94,053	20,182	10,990	48,971
	(% of Formal)	54.0	11.6	6.3	28.1
	Informal	111,773	2,232,216	151,792	1,531,660
	(% of Informal)	2.8	55.4	3.8	38.0
	Mix	10,522	3,344	7,219	8,023
	(% of Mix)	36.1	11.5	24.8	27.6
2019 Origin	Formal	197,196	54,785	20,666	108,369
	(% of Formal)	51.8	14.4	5.4	28.4
	Informal	84,059	2,547,652	62,012	1,473,395
	(% of Informal)	2.0	61.1	1.5	35.4
	Mix	48,735	83,256	57,455	62,577
	(% of Mix)	19.3	33.0	22.8	24.8

Note: Formal = 100% of workers with SS, Informal = 0% of workers with SS, 0% < Mix < 100% of workers with SS.

Back

Transition matrix, IMSS employer number defines informal

Transition matrix 2019-2024 (based on IMSS number)

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	84,146	73,597	93,991
	(% of Formal)	33.4	29.2	37.3
	Informal	142,684	2,261,091	1,331,140
	(% of Informal)	3.8	60.5	35.6

Note: It excludes the 813,508 (17%) missing values to the question about IMSS number.

Back

Transition matrix, RFC tax number defines informal

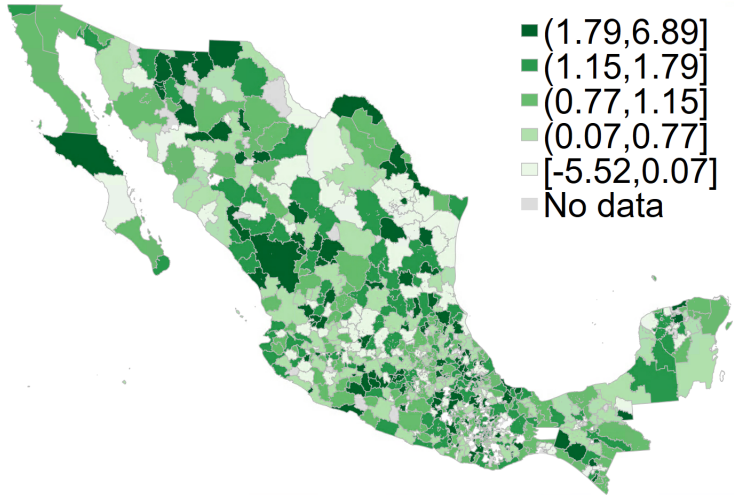
Transition matrix 2019-2024 (based on RFC number)

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	749,911	284,018	475,078
	(% of Formal)	49.7	18.8	31.5
	Informal	424,885	1,439,976	1,162,103
	(% of informal)	14.0	47.6	38.4

Note: It excludes the 264,186 (5%) missing values to the question about IMSS number.

Back

Five year credit change across markets



Back

Balance test

	Lagged 5 years		Lagged 10 years	
Predetermined variable	$\widehat{\beta}$	Std. error	$\widehat{\beta}$	Std. error
Employment share — Manufacturing	0.040	(0.026)	0.039	(0.027)
Employment share — Commerce	−0.008	(0.011)	−0.012	(0.012)
Bank credit share — Manufacturing	−0.007	(0.031)	0.027	(0.053)
Bank credit share — Commerce	0.010	(0.034)	−0.057	(0.061)
Credit-weighted average interest rate	0.323	(0.256)	0.756	(0.804)
Log number of banks	−0.039	(0.072)	−0.072	(0.069)
Log number of loans	0.205	(0.314)	−0.034	(0.315)
Observations (range)	1,204–1,360		770–938	
Markets (range)	521–545		459–517	

Each entry: separate regression of the predetermined characteristic on the standardized instrument, with year FE and coverage × year FE, weighted by baseline credit. S.e. clustered by market. None of the fourteen coefficients is distinguishable from zero at the 10 percent level.

Back

OLS and reduced-form estimates

Dependent variable (log)	**OLS on credit growth** $\widehat{\beta}$ (Std. error)	**Reduced form on** $\hat{Z}$ $\widehat{\beta}$ (Std. error)	N
Panel A: Stock and gross flows of firms			
Number of existing firms — Total	0.007 (0.005)	0.004 (0.012)	1,394
Number of existing firms — Formal	0.025** (0.012)	0.056*** (0.016)	1,389
Number of existing firms — Informal	0.006 (0.006)	−0.002 (0.013)	1,394
Firm entry — Total	0.017* (0.010)	0.017 (0.022)	996
Firm entry — Formal	0.033** (0.016)	0.075** (0.030)	979
Firm entry — Informal	0.017 (0.011)	0.013 (0.023)	996
Firm exit — Total	0.031*** (0.008)	0.055*** (0.011)	996
Firm exit — Formal	0.025** (0.011)	0.043* (0.025)	965
Firm exit — Informal	0.034*** (0.009)	0.062*** (0.011)	996
Panel B: Formality transitions			
Informal–Formal	0.031* (0.017)	0.059** (0.027)	957
Formal–Informal	−0.010 (0.013)	−0.044* (0.026)	958
Formal–Formal	0.027** (0.014)	0.045*** (0.016)	970
Informal–Informal	0.004 (0.006)	−0.007 (0.013)	996

OLS of the five-year log change on $\Delta_5 \ln(\text{credit})$ and reduced form on the standardized LOO instrument, same controls and weights; s.e. clustered by market.

▸ Back

Without three biggest metropolitan areas

Dependent variable	$\Delta_5 \widehat{\ln(\text{credit})}$	*N*
Panel A: Stock and gross flows of firms		
Log existing firms — Total	−0.003 (0.006)	1,385
Log existing firms — Formal	0.049*** (0.014)	1,380
Log existing firms — Informal	−0.009 (0.007)	1,385
Log firm entry — Total	0.018 (0.015)	990
Log firm entry — Formal	0.067*** (0.019)	973
Log firm entry — Informal	0.014 (0.015)	990
Log firm exit — Total	0.041** (0.018)	990
Log firm exit — Formal	0.067*** (0.018)	959
Log firm exit — Informal	0.037** (0.018)	990
Panel B: Formality transitions		
Formal–Formal	0.055*** (0.009)	964
Informal–Formal	0.014 (0.020)	951
Formal–Informal	−0.049** (0.023)	952
Informal–Informal	−0.025*** (0.006)	990

Excludes Mexico City, Monterrey, and Guadalajara (nine of the 1,394 market-window observations). AKM exposure-robust s.e.; first-stage F between 82 and 102.

Back

Pre-trend control

Dependent variable	**Main specification** $\widehat{\beta}$ (Std. error)	**With pre-trend control** $\widehat{\beta}$ (Std. error)
Log existing firms — Total	0.004 (0.005)	0.003 (0.005)
Log existing firms — Formal	0.050*** (0.013)	0.048*** (0.013)
Log existing firms — Informal	−0.002 (0.006)	−0.002 (0.006)
Formal workers (IMSS)	0.020*** (0.005)	0.019*** (0.006)
Formal wage bill (IMSS)	0.024** (0.009)	0.020** (0.010)
Mean daily wage (IMSS)	−0.012*** (0.004)	−0.008** (0.003)
Avg. value added	0.029* (0.015)	−0.031* (0.018)
Total value added	0.032* (0.017)	−0.045** (0.019)
Formal value added	0.028 (0.031)	−0.109*** (0.029)
Informal value added	−0.049** (0.021)	−0.071*** (0.014)

The pre-trend columns add the dependent variable's own lagged five-year change as a control (933–983 observations in 527–543 markets, against 1,386–1,394 in 552–556). AKM s.e. The firm margins hold; the aggregate and formal value-added coefficients reverse sign while the informal decline strengthens.

▸ Back

Placebo test

Pre-period outcome ($\Delta_5 Y_{t-5}$, log)	$\Delta_5 \widehat{\ln(\text{credit})}$	1st-stage F	N
Panel A: Stock and gross flows of firms			
Log existing firms — Total	0.016*** (0.005)	290.11	983
Log existing firms — Formal	0.016** (0.008)	290.06	979
Log existing firms — Informal	0.019*** (0.006)	290.11	983
Log firm entry — Total	0.034 (0.114)	1.89	514
Log firm entry — Formal	−0.078 (0.296)	1.89	508
Log firm entry — Informal	0.048 (0.095)	1.89	514
Log firm exit — Total	−0.043 (0.120)	1.89	514
Log firm exit — Formal	0.146 (0.368)	1.89	495
Log firm exit — Informal	−0.081 (0.108)	1.89	514
Panel B: Formality transitions			
Formal–Formal	−0.017 (0.077)	1.59	488
Informal–Formal	−0.112 (0.204)	1.89	496
Formal–Informal	0.075 (0.197)	1.89	485
Informal–Informal	−0.047 (0.127)	1.89	514

Stock rows use the full panel ($F = 290$); lagged flow changes exist only in one census window ($F = 1.9$, uninformative by construction). AKM s.e.

Back

Dropping the largest bank

Dependent variable	$\Delta_5 \widehat{\ln(\text{credit})}$	*N*
Panel A: Stock and gross flows of firms		
Log existing firms — Total	0.001 (0.005)	1,394
Log existing firms — Formal	0.046*** (0.010)	1,389
Log existing firms — Informal	−0.005 (0.007)	1,394
Log firm entry — Total	0.006 (0.011)	996
Log firm entry — Formal	0.056*** (0.012)	979
Log firm entry — Informal	0.003 (0.012)	996
Log firm exit — Total	0.044*** (0.015)	996
Log firm exit — Formal	−0.001 (0.016)	965
Log firm exit — Informal	0.059*** (0.015)	996
Panel B: Formality transitions		
Formal–Formal	0.028*** (0.009)	970
Informal–Formal	0.088*** (0.018)	957
Formal–Informal	−0.074*** (0.022)	958
Informal–Informal	0.000 (0.004)	996

Instrument rebuilt excluding the bank with the largest share of aggregate baseline credit (shocks and exposure shares removed). First-stage F between 34.95 and 35.13. AKM s.e.

Back

Conventional non-LOO estimates

Dependent variable	$\Delta_5 \widehat{\ln(\text{credit})}$	*N*
Panel A: Stock and gross flows of firms		
Log existing firms — Total	0.002 (0.005)	1,394
Log existing firms — Formal	0.035*** (0.007)	1,389
Log existing firms — Informal	−0.002 (0.006)	1,394
Log firm entry — Total	0.007 (0.009)	996
Log firm entry — Formal	0.043*** (0.009)	979
Log firm entry — Informal	0.007 (0.010)	996
Log firm exit — Total	0.049*** (0.009)	996
Log firm exit — Formal	0.021 (0.018)	965
Log firm exit — Informal	0.059*** (0.008)	996
Panel B: Formality transitions		
Formal–Formal	0.026*** (0.005)	970
Informal–Formal	0.049*** (0.019)	957
Formal–Informal	−0.043*** (0.012)	958
Informal–Informal	−0.003 (0.003)	996

Non-leave-one-out shift-share (bank shocks estimated including each market's own loans). Samples identical to the LOO regressions. First-stage F between 245 and 258. AKM s.e.

Back

Bank level estimation

Dependent variable	$\Delta_5 \widehat{\ln(\text{credit})}$	*N*
Panel A: Stock and gross flows of firms		
Log existing firms — Total	−0.004 (0.005)	131
Log existing firms — Formal	0.036*** (0.008)	131
Log existing firms — Informal	−0.009* (0.005)	131
Log firm entry — Total	−0.001 (0.008)	105
Log firm entry — Formal	0.045*** (0.011)	105
Log firm entry — Informal	−0.004 (0.009)	105
Log firm exit — Total	0.058*** (0.011)	105
Log firm exit — Formal	0.020 (0.016)	105
Log firm exit — Informal	0.070*** (0.010)	105
Panel B: Formality transitions		
Formal–Formal	0.021*** (0.008)	105
Informal–Formal	0.068*** (0.013)	105
Formal–Informal	−0.077*** (0.015)	105
Informal–Informal	−0.003 (0.004)	105

Bank-year level (Khwaja-Mian-style identification following Degryse et al., 2019); outcomes aggregated with the bank's baseline exposure shares. S.e. clustered by bank; first-stage F between 30.40 and 34.91.

Back

Different coverages

	Coverage ≥ 0			Coverage ≥ 0.8		
Dependent variable	$\hat{\beta}$ (Std. error)	*N*	*M*	$\hat{\beta}$ (Std. error)	*N*	*M*
Panel A: Stock and gross flows of firms						
Log existing firms — Total	0.006 (0.005)	1,520	568	0.003 (0.005)	1,218	541
Log existing firms — Formal	0.053^{***} (0.013)	1,514	564	0.045^{***} (0.012)	1,213	537
Log existing firms — Informal	−0.000 (0.005)	1,520	568	−0.001 (0.006)	1,218	541
Log firm entry — Total	0.016^{*} (0.009)	1,092	568	0.012 (0.009)	862	527
Log firm entry — Formal	0.068^{***} (0.017)	1,071	555	0.059^{***} (0.016)	846	516
Log firm entry — Informal	0.013 (0.009)	1,091	568	0.010 (0.010)	862	527
Log firm exit — Total	0.051^{***} (0.011)	1,092	568	0.047^{***} (0.012)	862	527
Log firm exit — Formal	0.037^{***} (0.014)	1,049	549	0.031^{**} (0.013)	834	510
Log firm exit — Informal	0.057^{***} (0.012)	1,092	568	0.056^{***} (0.013)	862	527
Panel B: Formality transitions						
Formal–Formal	0.038^{***} (0.008)	1,051	553	0.034^{***} (0.007)	837	513
Informal–Formal	0.064^{***} (0.018)	1,042	540	0.051^{***} (0.017)	827	500
Formal–Informal	-0.047^{***} (0.015)	1,040	541	-0.035^{**} (0.014)	829	505
Informal–Informal	−0.004 (0.004)	1,091	568	−0.004 (0.004)	862	527

AKM exposure-robust s.e.; first-stage F between 269 and 293 (left) and 288 and 325 (right). The main specification uses coverage ≥ 0.5.

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Operating value

Operating value

$$W_s(n, \hat{x}) = \max_{k'} \Big\{ \underbrace{n - k' - \Phi_s(a)}_{\text{dividend today}} + \beta \Big[\underbrace{(1-\delta)k'}_{\text{scrap}} + (1-\lambda_s) \underbrace{(\Pi_s(\hat{x}, k') - C_s)}_{\text{produce if alive}} \Big] \Big\}$$

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The formality margin: single crossing

For origin o (entrant, i-incumbent, f-incumbent) with funds n_o, on the borrowing region the formal-vs-informal comparison is *affine* in $\hat{x}^{1/(1-\gamma)}$:

$$\Delta_o(\hat{x}) = A\,\hat{x}^{\frac{1}{1-\gamma}} - B\,n_o - C_o, \qquad B = r_i - r_f > 0.$$

Assumption (finance-driven formality): $(R_f/R_i)^{\gamma} < T_\lambda \iff A > 0$: the financing advantage outweighs the fiscal cost (adjusted by survival, $T_\lambda = T \cdot \frac{1-\lambda_f}{1-\lambda_i}$).

$$\Rightarrow \quad \boxed{\hat{x}_o^F(n_o) = \left[\frac{B\,n_o + C_o}{A}\right]^{1-\gamma}, \qquad \frac{\partial \hat{x}_o^F}{\partial n_o} > 0}$$

Sorting result: formal firms are those with *high growth needs relative to internal wealth*.

- Conditional on expected productivity, *richer* firms are *less* likely to formalize, self-finance substitutes for the financial advantage of formality.

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Exit and entry threshold, four regimes

Every margin solves $G_i(\hat{x}, \tilde{n}) = m$ (continuation vs. consuming the outside value; the informal option is the relevant alternative at low $\hat{x}$):

$$\chi_s(\tilde{n}; m) = \begin{cases} (m/A_s^0)^{1-\gamma} & \text{Region 1, independent of finance} \\ \dfrac{m + R^0\tilde{n}}{M_s\tilde{n}^\gamma} & \text{Region 2} \\ \left[\dfrac{m - r_s\tilde{n}}{A_s}\right]^{1-\gamma} & \text{Region 3} \\ (1-b_s)^\gamma \dfrac{m + \tilde{n}\frac{R^0 + r_s b_s}{1-b_s}}{M_s\tilde{n}^\gamma} & \text{Region 4} \end{cases}$$

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